

PREPARED FOR

## Outkast Designs

test ca z4

36.95882, -122.03836

Section 23, T11S, R2W, Mount Diablo Meridian

✓ CORS Elev Delta: 620 ft below nearest CORS station (P534, 13.1 mi) – Good baseline geometry.

| COORDINATES                                    | ELEVATION                    | STATE PLANE                                                                              | FEMA ZONE                    | DECLINATION                |
|------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------|------------------------------|----------------------------|
| <b>36.9588°N</b><br><b>122.0384°W</b><br>WGS84 | <b>52 ft</b><br>NAVD88 · MSL | <b>CA 3 (0403) ·</b><br><b>N 1,811,174.36</b><br><b>E 6,112,182.90</b><br>(US ft, NAD83) | <b>Zone X</b><br>06087C0333F | <b>12.6° E</b><br>WMM 2026 |

APN 004-174-07 | ZONING CN | ACRES 0.108 ac

BEST GNSS WINDOW Thursday Jul 16 · 3:00 PM – 7:00 PM | PDOP 0.97 · 33 sv | NEAREST CORS  
P534 · 13.1 mi ✓

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VIEW FIELD MAP ↗

SITE SATELLITE ↗

EMERGENCY MAP ↗

SCAN FOR KMZ  
Site file · Google Earth  
Download KMZ

ASSESSOR-RECORDER Santa Cruz County Assessor-Recorder · 781 Ocean St. Santa Cruz, CA 95060

NEAREST CORS P534 · 13.1 mi

NEAREST BN GU1954 · 0.2 mi

GENERATED July 16, 2026

SYSTEM TopoScout · info@toposcout.com

## INTERACTIVE REPORT

This report includes clickable links and QR codes throughout – tap or scan them to open maps, official data sources, and field tools directly. Your site is also available as a drafting-ready CAD (DXF) file with georeferenced aerials. Add it anytime at [toposcout.com](https://toposcout.com).

## SITE & SIGNAL REPORT

### COORDINATES

**36.9588, -122.0384**

WGS84 / NAD83

### ELEVATION

**52 ft**

Above MSL (approx)

### MAGNETIC DECLINATION

**12.6° E**

Compass reads 12.6° too far W

### FEMA FLOOD ZONE

**X**

AREA OF MINIMAL FLOOD HAZARD · Panel 06087C0333F

## IONOSPHERIC CONDITIONS

### PEAK KP INDEX

**4.7**

### AVG KP

**2.3**

### SOLAR FLUX F10.7

**138** sfu

### TEC (VTEC) ▲ DEGRADED

**18.7** TECU

-5.5 TECU vs median

## ASSESSMENT

**MODERATE ACTIVITY**

Peak Kp 4.7 (avg 2.3). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

## GNSS / DOP SUMMARY

| DATE        | AVG PDOP | BEST 4-HR WINDOW   | AVG SATS | RATING |
|-------------|----------|--------------------|----------|--------|
| Thu, Jul 16 | 1.01     | 3:00 PM – 7:00 PM  | 33       | Good   |
| Fri, Jul 17 | 1.00     | 3:00 PM – 7:00 PM  | 33       | Good   |
| Sat, Jul 18 | 0.97     | 3:00 PM – 7:00 PM  | 35       | Ideal  |
| Sun, Jul 19 | 0.99     | 2:00 PM – 6:00 PM  | 35       | Ideal  |
| Mon, Jul 20 | 0.98     | 6:00 AM – 10:00 AM | 35       | Ideal  |
| Tue, Jul 21 | 0.99     | 3:00 PM – 7:00 PM  | 33       | Ideal  |
| Wed, Jul 22 | 1.01     | 4:00 PM – 8:00 PM  | 34       | Good   |

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation  
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

Days analyzed: 7   Recommended work days: 5 of 7   Best PDOP: 0.97

►

RECOMMENDED WINDOW

Mon, Jul 20

6:00 AM – 10:00 AM

Avg wind 3 mph · Gusts 3 mph

Ideal conditions — 4-hr static occupation

ALTERNATE WINDOW

Tue, Jul 21

4:00 PM – 8:00 PM

Avg wind 5 mph · Gusts 5 mph

Ideal conditions — 4-hr static occupation

🔥

WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 16, 2026) · Conditions may change – verify before fieldwork

📶

GPS CONSTELLATION HEALTH

⚠️ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

MULTIPATH RISK

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–1.1 across survey windows

PDOP remains ≤ 1.1 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

| Point  | Cover Type  | Obstruction | Multipath |
|--------|-------------|-------------|-----------|
| Center | Dwarf Scrub | Low         | Low       |
| North  | Dwarf Scrub | Low         | Low       |
| South  | Dwarf Scrub | Low         | Low       |
| East   | Dwarf Scrub | Low         | Low       |
| West   | Dwarf Scrub | Low         | Low       |

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 82°F

Low 55°F

Wind max 15 mph

Gusts 17 mph

Precip 0.00"

Sunrise 6:01 AM

Sunset 8:26 PM

New Moon 3%

GNSS: Avg PDOP **1.01** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.97, 33 sats avg)

| GNSS / DOP - HOURLY |      |  |       |        |
|---------------------|------|--|-------|--------|
|                     | PDOP |  | SATS  | RATING |
| 6:00 AM             | 0.93 |  | 32 sv | IDEAL  |
| 7:00 AM             | 0.99 |  | 32 sv | IDEAL  |
| 8:00 AM             | 0.87 |  | 36 sv | IDEAL  |
| 9:00 AM             | 1.20 |  | 27 sv | GOOD   |
| 10:00 AM            | 1.11 |  | 27 sv | GOOD   |
| 11:00 AM            | 1.05 |  | 27 sv | GOOD   |
| 12:00 PM            | 0.90 |  | 31 sv | IDEAL  |
| 1:00 PM             | 0.99 |  | 30 sv | IDEAL  |
| 2:00 PM             | 1.13 |  | 29 sv | GOOD   |
| 3:00 PM             | 0.95 |  | 33 sv | IDEAL  |
| 4:00 PM             | 0.95 |  | 35 sv | IDEAL  |
| 5:00 PM             | 0.90 |  | 35 sv | IDEAL  |
| 6:00 PM             | 1.08 |  | 29 sv | GOOD   |
| 7:00 PM             | 1.13 |  | 27 sv | GOOD   |



GO – GOOD CONDITIONS

Clear sky | Wind: 14.6 mph (gusts 17 mph) | Precip: 0" | Temp: 54.7°–81.8°F



DRONE WIND ANALYSIS



GO Peak: 14.6/17 mph WNW

\* Sunrise: 6:01 AM    ☾ Sunset: 8:26 PM    fly window clipped to daylight

✓ Best flight window: 7:00 AM – 9:00 PM (14 hrs)

🌀 Wind direction shifting — plan mission headings carefully

|       | SUSTAINED / GUSTS mph | ≠turbulence | DIR |
|-------|-----------------------|-------------|-----|
| 5 AM  | 2/2 mph               |             | NE  |
| 6 AM  | 3/3 mph               |             | WSW |
| 7 AM  | 2/3 mph               |             | SW  |
| 8 AM  | 1/1 mph               |             | SE  |
| 9 AM  | 4/4 mph               |             | SSE |
| 10 AM | 3/3 mph               |             | S   |
| 11 AM | 5/5 mph               |             | WSW |
| 12 PM | 6/6 mph               |             | W   |
| 1 PM  | 10/11 mph             |             | WNW |
| 2 PM  | 13/16 mph             |             | WNW |
| 3 PM  | 15/17 mph             |             | WNW |
| 4 PM  | 14/17 mph             |             | WNW |
| 5 PM  | 13/16 mph             |             | WNW |
| 6 PM  | 12/15 mph             |             | WNW |
| 7 PM  | 10/14 mph             |             | NW  |
| 8 PM  | 8/13 mph              |             | WNW |

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 73°F

Low 53°F

Wind max 14 mph

Gusts 17 mph

Precip 0.00"

Sunrise 6:02 AM

Sunset 8:26 PM

Waxing Crescent 8%

GNSS: Avg PDOP **1.00** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.93, 33 sats avg)

| GNSS / DOP - HOURLY |      |  |       |  |        |  |
|---------------------|------|--|-------|--|--------|--|
|                     | PDOP |  | SATS  |  | RATING |  |
| 6:00 AM             | 0.91 |  | 34 sv |  | IDEAL  |  |
| 7:00 AM             | 1.26 |  | 28 sv |  | GOOD   |  |
| 8:00 AM             | 0.91 |  | 35 sv |  | IDEAL  |  |
| 9:00 AM             | 0.93 |  | 33 sv |  | IDEAL  |  |
| 10:00 AM            | 0.99 |  | 30 sv |  | IDEAL  |  |
| 11:00 AM            | 0.96 |  | 28 sv |  | IDEAL  |  |
| 12:00 PM            | 0.92 |  | 31 sv |  | IDEAL  |  |
| 1:00 PM             | 1.12 |  | 27 sv |  | GOOD   |  |
| 2:00 PM             | 1.04 |  | 31 sv |  | GOOD   |  |
| 3:00 PM             | 0.94 |  | 34 sv |  | IDEAL  |  |
| 4:00 PM             | 0.93 |  | 35 sv |  | IDEAL  |  |
| 5:00 PM             | 0.95 |  | 33 sv |  | IDEAL  |  |
| 6:00 PM             | 0.91 |  | 31 sv |  | IDEAL  |  |
| 7:00 PM             | 1.17 |  | 25 sv |  | GOOD   |  |



GO – GOOD CONDITIONS

Partly cloudy | Wind: 14.2 mph (gusts 16.6 mph) | Precip: 0" | Temp: 53°–72.5°F



DRONE WIND ANALYSIS



GO Peak: 14.2/16.6 mph W

\* Sunrise: 6:02 AM    Sunset: 8:26 PM    fly window clipped to daylight

Best flight window: 7:00 AM – 9:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

|       | SUSTAINED / GUSTS mph | ≠turbulence | DIR |
|-------|-----------------------|-------------|-----|
| 5 AM  | 4/5 mph               |             | NW  |
| 6 AM  | 2/3 mph               |             | NNW |
| 7 AM  | 2/3 mph               |             | ENE |
| 8 AM  | 2/2 mph               |             | SE  |
| 9 AM  | 2/2 mph               |             | SSW |
| 10 AM | 4/5 mph               |             | WSW |
| 11 AM | 10/11 mph             |             | W   |
| 12 PM | 13/14 mph             |             | W   |
| 1 PM  | 13/15 mph             |             | W   |
| 2 PM  | 14/17 mph             |             | W   |
| 3 PM  | 13/14 mph             |             | W   |
| 4 PM  | 10/11 mph             |             | WSW |
| 5 PM  | 7/7 mph               |             | WSW |
| 6 PM  | 4/3 mph               |             | SSW |
| 7 PM  | 2/6 mph               |             | ESE |
| 8 PM  | 3/5 mph               |             | E   |

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY  
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107  
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗  
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#3

Saturday, Jul 18

WEEKEND

Clear sky

8 mph

GO

High 72°F

Low 57°F

Wind max 8 mph

Gusts 6 mph

Precip 0.00"

Sunrise 6:02 AM

Sunset 8:25 PM

Waxing Crescent 15%

GNSS: Avg PDOP 0.97 · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.92, 35 sats avg)

GNSS / DOP - HOURLY

|          | PDOP | SATS  | RATING |
|----------|------|-------|--------|
| 6:00 AM  | 0.91 | 34 sv | IDEAL  |
| 7:00 AM  | 1.02 | 32 sv | GOOD   |
| 8:00 AM  | 1.03 | 32 sv | GOOD   |
| 9:00 AM  | 0.94 | 33 sv | IDEAL  |
| 10:00 AM | 1.01 | 30 sv | GOOD   |
| 11:00 AM | 1.00 | 29 sv | GOOD   |
| 12:00 PM | 0.90 | 32 sv | IDEAL  |
| 1:00 PM  | 1.02 | 31 sv | GOOD   |
| 2:00 PM  | 0.94 | 34 sv | IDEAL  |
| 3:00 PM  | 0.88 | 36 sv | IDEAL  |
| 4:00 PM  | 0.93 | 36 sv | IDEAL  |
| 5:00 PM  | 0.99 | 32 sv | IDEAL  |
| 6:00 PM  | 0.89 | 35 sv | IDEAL  |
| 7:00 PM  | 1.11 | 30 sv | GOOD   |



GO - GOOD CONDITIONS

Clear sky | Wind: 8.2 mph (gusts 6.3 mph) | Precip: 0" | Temp: 56.5°–72.1°F



DRONE WIND ANALYSIS



GO Peak: 8.2/6.3 mph SSE

\* Sunrise: 6:02 AM    Sunset: 8:25 PM    fly window clipped to daylight



Best flight window: 7:00 AM – 9:00 PM (14 hrs)



Wind direction shifting — plan mission headings carefully

|       | SUSTAINED / GUSTS mph | ≤turbulence | DIR |
|-------|-----------------------|-------------|-----|
| 5 AM  | 4/3 mph               |             | SE  |
| 6 AM  | 4/3 mph               |             | SE  |
| 7 AM  | 3/3 mph               |             | SE  |
| 8 AM  | 3/2 mph               |             | SSE |
| 9 AM  | 4/2 mph               |             | S   |
| 10 AM | 5/3 mph               |             | SSW |
| 11 AM | 7/4 mph               |             | SSW |
| 12 PM | 8/4 mph               |             | SW  |
| 1 PM  | 8/5 mph               |             | SW  |
| 2 PM  | 8/6 mph               |             | SW  |
| 3 PM  | 7/6 mph               |             | SW  |
| 4 PM  | 7/6 mph               |             | SSW |
| 5 PM  | 6/6 mph               |             | SSW |
| 6 PM  | 4/3 mph               |             | SSW |
| 7 PM  | 2/3 mph               |             | SSE |
| 8 PM  | 3/5 mph               |             | ESE |

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#4

Sunday, Jul 19

Overcast

9 mph

GO

WEEKEND

High 70°F

Low 57°F

Wind max 9 mph

Gusts 8 mph

Precip 0.00"

Sunrise 6:03 AM

Sunset 8:25 PM

Waxing Crescent 23%

GNSS: Avg PDOP 0.99 · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.94, 35 sats avg)

| GNSS / DOP - HOURLY |      |  |       |  |        |
|---------------------|------|--|-------|--|--------|
|                     | PDOP |  | SATS  |  | RATING |
| 6:00 AM             | 0.96 |  | 34 sv |  | IDEAL  |
| 7:00 AM             | 1.04 |  | 29 sv |  | GOOD   |
| 8:00 AM             | 0.95 |  | 32 sv |  | IDEAL  |
| 9:00 AM             | 0.91 |  | 32 sv |  | IDEAL  |
| 10:00 AM            | 1.13 |  | 27 sv |  | GOOD   |
| 11:00 AM            | 1.14 |  | 25 sv |  | GOOD   |
| 12:00 PM            | 1.03 |  | 27 sv |  | GOOD   |
| 1:00 PM             | 0.93 |  | 31 sv |  | IDEAL  |
| 2:00 PM             | 0.91 |  | 35 sv |  | IDEAL  |
| 3:00 PM             | 1.02 |  | 32 sv |  | GOOD   |
| 4:00 PM             | 0.90 |  | 37 sv |  | IDEAL  |
| 5:00 PM             | 0.92 |  | 34 sv |  | IDEAL  |
| 6:00 PM             | 0.98 |  | 32 sv |  | IDEAL  |
| 7:00 PM             | 1.07 |  | 29 sv |  | GOOD   |

GO — GOOD CONDITIONS

Partly cloudy | Wind: 8.7 mph (gusts 8.1 mph) | Precip: 0" | Temp: 57.4°–70.3°F

DRONE WIND ANALYSIS

GO Peak: 8.7/8.1 mph SSW

\* Sunrise: 6:03 AM    Sunset: 8:25 PM    fly window clipped to daylight

Best flight window: 7:00 AM – 9:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

|       | SUSTAINED / GUSTS mph | ≠turbulence | DIR |
|-------|-----------------------|-------------|-----|
| 5 AM  | 2/2 mph               |             | SSE |
| 6 AM  | 2/2 mph               |             | SSE |
| 7 AM  | 3/2 mph               |             | SE  |
| 8 AM  | 5/3 mph               |             | SSE |
| 9 AM  | 4/2 mph               |             | SSE |
| 10 AM | 4/2 mph               |             | SSW |
| 11 AM | 7/3 mph               |             | SSW |
| 12 PM | 8/4 mph               |             | SW  |
| 1 PM  | 7/5 mph               |             | SSW |
| 2 PM  | 8/6 mph               |             | SW  |
| 3 PM  | 9/8 mph               |             | SW  |
| 4 PM  | 7/7 mph               |             | SW  |
| 5 PM  | 7/7 mph               |             | SW  |
| 6 PM  | 7/6 mph               |             | SSW |
| 7 PM  | 4/5 mph               |             | SSW |
| 8 PM  | 2/3 mph               |             | SW  |

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY  
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107  
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)  
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 71°F

Low 59°F

Wind max 8 mph

Gusts 6 mph

Precip 0.00"

Sunrise 6:04 AM

Sunset 8:24 PM

Waxing Crescent 33%

GNSS: Avg PDOP **0.98** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.92, 35 sats avg)

| GNSS / DOP - HOURLY |      |       |        |
|---------------------|------|-------|--------|
|                     | PDOP | SATS  | RATING |
| 6:00 AM             | 0.88 | 35 sv | IDEAL  |
| 7:00 AM             | 0.97 | 34 sv | IDEAL  |
| 8:00 AM             | 0.88 | 37 sv | IDEAL  |
| 9:00 AM             | 0.95 | 32 sv | IDEAL  |
| 10:00 AM            | 0.94 | 29 sv | IDEAL  |
| 11:00 AM            | 1.11 | 25 sv | GOOD   |
| 12:00 PM            | 1.00 | 28 sv | GOOD   |
| 1:00 PM             | 1.02 | 27 sv | GOOD   |
| 2:00 PM             | 0.99 | 33 sv | IDEAL  |
| 3:00 PM             | 0.98 | 34 sv | IDEAL  |
| 4:00 PM             | 0.94 | 33 sv | IDEAL  |
| 5:00 PM             | 0.97 | 32 sv | IDEAL  |
| 6:00 PM             | 0.92 | 34 sv | IDEAL  |
| 7:00 PM             | 1.19 | 28 sv | GOOD   |

**GO – GOOD CONDITIONS**  
Partly cloudy | Wind: 7.8 mph (gusts 6 mph) | Precip: 0" | Temp: 58.5°–71.1°F

**DRONE WIND ANALYSIS** ✓ **GO** Peak: 7.8/6 mph S

\* Sunrise: 6:04 AM    ☾ Sunset: 8:24 PM    fly window clipped to daylight

☒ Best flight window: 7:00 AM – 9:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

|       | SUSTAINED / GUSTS mph | ≠turbulence | DIR |
|-------|-----------------------|-------------|-----|
| 5 AM  | 3/2 mph               |             | ESE |
| 6 AM  | 4/3 mph               |             | ESE |
| 7 AM  | 2/2 mph               |             | SE  |
| 8 AM  | 3/3 mph               |             | SSW |
| 9 AM  | 3/2 mph               |             | SE  |
| 10 AM | 8/5 mph               |             | SW  |
| 11 AM | 7/4 mph               |             | SW  |
| 12 PM | 7/5 mph               |             | SW  |
| 1 PM  | 7/6 mph               |             | SW  |
| 2 PM  | 7/6 mph               |             | WSW |
| 3 PM  | 6/6 mph               |             | SW  |
| 4 PM  | 7/6 mph               |             | SW  |
| 5 PM  | 6/5 mph               |             | SSW |
| 6 PM  | 5/4 mph               |             | SSW |
| 7 PM  | 3/3 mph               |             | SSW |
| 8 PM  | 2/2 mph               |             | S   |

**BEST WINDOW**  
**7:00 AM – 9:00 PM**  
*Ideal conditions — calm winds, clean operations all day*

AIRSPACE **CLASS A** ▶ FREE TO FLY  
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107  
→ TFR STATUS ☒ **NO TFRs WITHIN 30 MI** [Check B4UFLY](#) ↗  
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 77°F

Low 60°F

Wind max 8 mph

Gusts 5 mph

Precip 0.00"

Sunrise 6:05 AM

Sunset 8:24 PM

Waxing Crescent 43%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.94, 33 sats avg)

| GNSS / DOP - HOURLY |      |  |       |  |        |
|---------------------|------|--|-------|--|--------|
|                     | PDOP |  | SATS  |  | RATING |
| 6:00 AM             | 0.95 |  | 32 sv |  | IDEAL  |
| 7:00 AM             | 0.97 |  | 34 sv |  | IDEAL  |
| 8:00 AM             | 0.89 |  | 35 sv |  | IDEAL  |
| 9:00 AM             | 1.13 |  | 30 sv |  | GOOD   |
| 10:00 AM            | 1.01 |  | 31 sv |  | GOOD   |
| 11:00 AM            | 1.07 |  | 28 sv |  | GOOD   |
| 12:00 PM            | 0.96 |  | 30 sv |  | IDEAL  |
| 1:00 PM             | 1.06 |  | 28 sv |  | GOOD   |
| 2:00 PM             | 1.03 |  | 30 sv |  | GOOD   |
| 3:00 PM             | 0.94 |  | 33 sv |  | IDEAL  |
| 4:00 PM             | 0.94 |  | 36 sv |  | IDEAL  |
| 5:00 PM             | 0.96 |  | 32 sv |  | IDEAL  |
| 6:00 PM             | 0.93 |  | 31 sv |  | IDEAL  |
| 7:00 PM             | 1.03 |  | 29 sv |  | GOOD   |

**GO — GOOD CONDITIONS**  
Partly cloudy | Wind: 7.5 mph (gusts 5.1 mph) | Precip: 0" | Temp: 60°–76.6°F

**DRONE WIND ANALYSIS** ✓ **GO** Peak: 7.5/5.1 mph SSE

\* Sunrise: 6:05 AM    ☾ Sunset: 8:24 PM    fly window clipped to daylight

✓ Best flight window: 7:00 AM – 9:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

|       | SUSTAINED / GUSTS mph | ≈turbulence | DIR |
|-------|-----------------------|-------------|-----|
| 5 AM  | 1/1 mph               |             | SW  |
| 6 AM  | 2/2 mph               |             | S   |
| 7 AM  | 4/3 mph               |             | SSE |
| 8 AM  | 5/3 mph               |             | SSE |
| 9 AM  | 5/3 mph               |             | S   |
| 10 AM | 5/3 mph               |             | S   |
| 11 AM | 5/3 mph               |             | SSW |
| 12 PM | 6/4 mph               |             | SSW |
| 1 PM  | 7/4 mph               |             | SW  |
| 2 PM  | 8/5 mph               |             | SW  |
| 3 PM  | 7/4 mph               |             | SSW |
| 4 PM  | 5/4 mph               |             | SSW |
| 5 PM  | 5/4 mph               |             | S   |
| 6 PM  | 4/4 mph               |             | SSE |
| 7 PM  | 4/5 mph               |             | SSE |
| 8 PM  | 4/5 mph               |             | SE  |

**BEST WINDOW**  
**7:00 AM – 9:00 PM**  
*Ideal conditions — calm winds, clean operations all day*

AIRSPACE **CLASS A** ▶ FREE TO FLY  
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107  
→ TFR STATUS ✓ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)  
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 76°F

Low 60°F

Wind max 9 mph

Gusts 6 mph

Precip 0.00"

Sunrise 6:05 AM

Sunset 8:23 PM

First Quarter 53%

GNSS: Avg PDOP **1.01** · Best 4-hr window 4:00 PM – 8:00 PM (PDOP 0.91, 34 sats avg)

| GNSS / DOP - HOURLY |      |       |        |
|---------------------|------|-------|--------|
|                     | PDOP | SATS  | RATING |
| 6:00 AM             | 1.15 | 29 sv | GOOD   |
| 7:00 AM             | 1.05 | 30 sv | GOOD   |
| 8:00 AM             | 0.98 | 34 sv | IDEAL  |
| 9:00 AM             | 1.05 | 30 sv | GOOD   |
| 10:00 AM            | 1.02 | 29 sv | GOOD   |
| 11:00 AM            | 1.17 | 25 sv | GOOD   |
| 12:00 PM            | 0.97 | 28 sv | IDEAL  |
| 1:00 PM             | 1.01 | 29 sv | GOOD   |
| 2:00 PM             | 0.98 | 34 sv | IDEAL  |
| 3:00 PM             | 1.17 | 33 sv | GOOD   |
| 4:00 PM             | 0.87 | 36 sv | IDEAL  |
| 5:00 PM             | 0.89 | 34 sv | IDEAL  |
| 6:00 PM             | 0.92 | 32 sv | IDEAL  |
| 7:00 PM             | 0.94 | 32 sv | IDEAL  |

GO — GOOD CONDITIONS

Partly cloudy | Wind: 8.6 mph (gusts 5.8 mph) | Precip: 0" | Temp: 60.1°–75.7°F

DRONE WIND ANALYSIS

GO Peak: 8.6/5.8 mph SSE

\* Sunrise: 6:05 AM    ☾ Sunset: 8:23 PM    fly window clipped to daylight

Best flight window: 7:00 AM – 9:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

|       | SUSTAINED / GUSTS mph | ≈turbulence | DIR |
|-------|-----------------------|-------------|-----|
| 5 AM  | 3/3 mph               |             | ESE |
| 6 AM  | 3/2 mph               |             | ESE |
| 7 AM  | 3/2 mph               |             | SE  |
| 8 AM  | 3/2 mph               |             | SSE |
| 9 AM  | 3/3 mph               |             | S   |
| 10 AM | 5/3 mph               |             | SSW |
| 11 AM | 7/4 mph               |             | SSW |
| 12 PM | 8/5 mph               |             | SW  |
| 1 PM  | 9/5 mph               |             | SW  |
| 2 PM  | 8/6 mph               |             | SW  |
| 3 PM  | 6/6 mph               |             | SSW |
| 4 PM  | 5/5 mph               |             | SSE |
| 5 PM  | 6/5 mph               |             | SE  |
| 6 PM  | 7/6 mph               |             | SE  |
| 7 PM  | 7/6 mph               |             | ESE |
| 8 PM  | 7/6 mph               |             | ESE |

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

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→ TFR STATUS NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)  
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

**FIELD PLANNING REPORT**

GNSS SURVEY OPERATIONS — SANTA CRUZ COUNTY, CA

Report Generated: Thursday, 16 July 2026 | Survey Start: Thursday, 16 July 2026

=====

TODAY (THURSDAY, JULY 16) — DEPLOYMENT STATUS

=====

RECOMMENDATION: GO — CONDITIONS FAVORABLE

PDOP & GEOMETRY: Excellent. Avg PDOP 1.01, peak window 3:00 PM–7:00 PM delivers PDOP 0.97 with 33 satellites (GPS+GLONASS+Galileo+BeiDou).

WEATHER: Clear skies, H 82°F, wind max 15 mph, zero precipitation. Optimal for setup and crew logistics.

IONOSPHERE: Moderate geomagnetic activity (Kp avg 2.3, peak 4.7). TEC 18.7 TECU, 5.5 TECU below 30-day median. Ionospheric delay manageable; static sessions should use standard occupation times (1–2 hr minimum).

SITE CONDITIONS: Dwarf scrub terrain, low sky obstruction, low multipath risk. Recommended elevation mask 13°. No wildfire threat.

SATELLITE HEALTH: PRN 11 unhealthy (auto-excluded by receivers). PRN 26 maintenance window Fri 00:35–12:35 UTC (informational; does not degrade Thu). Multi-constellation availability maintains strong geometry.

VERDICT: Deploy today if personnel and equipment are ready. Excellent conditions for baseline static work afternoon into evening.

=====

MONDAY CALL — CRITICAL MOBILIZATION DECISION

=====

RECOMMENDATION: GO — PROCEED WITH MOBILIZATION

PDOP & GEOMETRY: Excellent. Avg PDOP 0.98, peak window 6:00 AM–10:00 AM yields PDOP 0.92 with 35 satellites. Morning geometry is strongest of the week.

WEATHER: Overcast, H 71°F, wind max 8 mph, zero precipitation. No impact on GNSS quality.

IONOSPHERE: Kp activity moderating; no deterioration expected by Monday. TEC anomaly recovering toward climatological values.

VERDICT: Crews mobilize as planned. Monday morning (6:00 AM–10:00 AM) is one of the highest-confidence windows of the week. No delays justified.

=====

WEEK RANKING — ALL 7 DAYS (1=BEST, 7=WORST)

=====

RANK 1: MONDAY, JULY 20

PDOP 0.98 avg, best window 6:00 AM–10:00 AM (0.92, 35 sats). Overcast, benign wind. Peak geometry of week; early-morning daylight leverage.

RANK 2: SATURDAY, JULY 18 [WEEKEND]

PDOP 0.97 avg, best window 3:00 PM–7:00 PM (0.92, 35 sats). Clear skies, lightest winds (8 mph). Highest satellite count. Weather excellent. WEEKEND FLAG.

RANK 3: FRIDAY, JULY 17

PDOP 1.00 avg, best window 3:00 PM–7:00 PM (0.93, 33 sats). Overcast; zero precip. PRN 26 maintenance 00:35–12:35 UTC (morning hours only; afternoon window unaffected). Good geometry, reliable conditions.

**RANK 4: TUESDAY, JULY 21**

PDOP 0.99 avg, best window 3:00 PM–7:00 PM (0.94, 33 sats). Overcast; warm (77°F), calm winds. Acceptable geometry; mid-week operational rhythm.

**RANK 5: THURSDAY, JULY 16 (TODAY)**

PDOP 1.01 avg, best window 3:00 PM–7:00 PM (0.97, 33 sats). Clear skies, warmer, wind 15 mph (highest of week). Excellent for immediate deployment; slightly elevated wind.

**RANK 6: SUNDAY, JULY 19 [WEEKEND]**

PDOP 0.99 avg, best window 2:00 PM–6:00 PM (0.94, 35 sats). Overcast, calm winds. Geometry solid but WEEKEND FLAG and mid-afternoon start limit full-day utility.

**RANK 7: WEDNESDAY, JULY 22**

PDOP 1.01 avg, best window 4:00 PM–8:00 PM (0.91, 34 sats). Overcast, late-afternoon window compresses daylight. Acceptable but end-of-week fatigue and scheduling constraints.

**TOP-5 RECOMMENDED WORK DAYS:** Monday (20th), Saturday (18th), Friday (17th), Tuesday (21st), Thursday (16th).

=====

**BEST STATIC WINDOW — WEEK OVERVIEW**

=====

**OPTIMAL 4-HOUR STATIC SESSION:**

**WINDOW:** Monday, July 20 | 6:00 AM – 10:00 AM (Local Time)

**RATIONALE:** Peak geometry of entire week (PDOP 0.92, 35 satellites). Morning light for crew safety and site logistics. No weather degradation. Moderate ionosphere; occupation time 1–2 hours standard protocol meets TEC conditions. Delivers highest-confidence baseline to start week operations.

**ALTERNATIVE** (if weather delays Monday morning): Friday, July 17 | 3:00 PM – 7:00 PM (PDOP 0.93, 33 sats). Longer daylight buffer; overcast sky irrelevant for GNSS.

=====

**DRONE ASSESSMENT — DAY-BY-DAY AIRWORTHINESS**

=====

**THURSDAY, JULY 16:** FLY WITH CAUTION — Wind 15 mph elevated; otherwise clear. Proceed if platform rated; brief crews on gust gusts in open scrub terrain.

**FRIDAY, JULY 17:** FLY — Overcast, wind 14 mph. Acceptable. Standard pre-flight checks.

**SATURDAY, JULY 18:** FLY — Clear sky, wind 8 mph. Optimal conditions. **WEEKEND** — scheduling permitting, execute any aerial surveys today.

**SUNDAY, JULY 19:** FLY — Overcast, wind 9 mph. Acceptable.

**MONDAY, JULY 20:** FLY — Overcast, wind 8 mph. Acceptable; morning window best for early-light imagery.

**TUESDAY, JULY 21:** FLY — Overcast, wind 8 mph. Acceptable.

**WEDNESDAY, JULY 22:** FLY — Overcast, wind 9 mph. Acceptable; late afternoon may compress ideal light window.

**WILDFIRE & AQI:** No active fires within range. Regional air quality forecast not degraded. No aerial operations restrictions anticipated.

=====

**FIELD WARNINGS — CRITICAL SAFETY & OPERATIONAL NOTES**

=====

**SATELLITE CONSTELLATION:**

- PRN 11 (GPS) currently unhealthy (UNUSUFN). No return time published. Modern multi-constellation receivers auto-exclude; geometry remains excellent (35-satellite redundancy typical mid-morning). No action required.
- PRN 26 scheduled maintenance Fri 00:35–12:35 UTC. Friday afternoon static (3:00 PM–7:00 PM local, well after 12:35 UTC) unaffected. Informational only.
- PRN 3 informational maintenance windows (past July 9 start) noted. Do not affect this survey week.

**IONOSPHERIC HAZARD (MODERATE):**

- Kp index peaked 4.7 (moderate geomagnetic storm threshold). Average 2.3 over 7-day window.
- TEC 18.7 TECU, 5.5 TECU below median (favorable). However, quality flag non-zero; treat estimates conservatively.
- RECOMMENDATION: Extend static occupation times to 1.5–2 hours minimum, especially Thursday–Friday. Use post-processing ionospheric models (CODE, IGS final) for baseline reduction.

**WEATHER:**

- No precipitation forecast entire week. Terrain drainage (dwarf scrub, sandy/rocky) adequate.
- Wind: Thursday elevated (15 mph); all other days 8–9 mph. No storm threat.
- Temperatures 55–82°F. Standard field operations conditions; ensure crew hydration (high elevations, sun exposure).

**TERRAIN & SITE CONDITIONS:**

- Low sky obstruction (dwarf scrub, open terrain). Elevation mask 13° recommended. Multipath risk low.
- No wildfire proximity. Regional air quality clear.
- Typical Santa Cruz County coastal conditions: clear mornings, marine layer afternoon overcast (consistent with forecast). Plan critical work 6:00 AM–12:00 PM for clearest skies.

=====

**OVERALL RECOMMENDATION**

=====

**CREW CHIEF DIRECTIVE:**

Deploy today (Thursday) to establish base operations and acquire preliminary static data during 3:00 PM–7:00 PM window (PDOP 0.97, clear skies). Mobilize main crew for Monday morning start; Monday 6:00 AM–10:00 AM is the highest-confidence GNSS window of the week (PDOP 0.92, 35 satellites). Execute secondary or drone surveys Saturday (weather and geometry optimal); Friday afternoon and Tuesday afternoon are reliable alternatives if Monday operations extend. Extend static occupation times to 1.5–2 hours minimum due to moderate ionospheric activity; apply final ionospheric corrections during post-processing.

=====

**END OF REPORT**

**COUNTY RECORDING OFFICES**

RECORDER / REGISTER OF DEEDS

Santa Cruz County Assessor-Recorder  
701 Ocean St, Santa Cruz, CA 95060

ASSESSOR / TAX OFFICE

Santa Cruz County Assessor  
701 Ocean St, Santa Cruz, CA 95060

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

CORS STATIONS – WITHIN 30 MILES

| QR / LINK                                                                                  | STATION ID  | NAME / LOCATION  | DISTANCE |
|--------------------------------------------------------------------------------------------|-------------|------------------|----------|
| <br>NGS ↗ | <b>P534</b> | COOKEPEAK_CN2007 | 13.1 mi  |
| <br>NGS ↗ | <b>P210</b> | ELKHNSLGHNCN2005 | 19.6 mi  |
| <br>NGS ↗ | <b>P217</b> | LACROSSEDRCN2005 | 23.7 mi  |
| <br>NGS ↗ | <b>P277</b> | PIGEONPT__CN2007 | 24.3 mi  |
| <br>NGS ↗ | <b>P231</b> | HOPKINSSTNCN2006 | 24.4 mi  |
| <br>NGS ↗ | <b>PPT6</b> | PIGEON POINT 6   | 25.0 mi  |

NGS BENCHMARKS – WITHIN 10 MILES

| QR / LINK                                                                                          | PID           | DESIGNATION | ELEVATION               | ORDER   | DISTANCE |
|----------------------------------------------------------------------------------------------------|---------------|-------------|-------------------------|---------|----------|
| <br>Datasheet ↗  | <b>GU1954</b> | "V"         | 17.526m / 57.5ft NAVD88 | Order 1 | 0.2 mi   |
| <br>Datasheet ↗ | <b>GU1955</b> | "U"         | 18.570m / 60.9ft NAVD88 | Order   | 0.2 mi   |
| <br>Datasheet ↗ | <b>GU4240</b> | "M"         | 17.293m / 56.7ft NAVD88 | Order 1 | 0.4 mi   |
| <br>Datasheet ↗ | <b>GU1950</b> | "U"         | 3.505m / 11.5ft NAVD88  | Order   | 0.7 mi   |
| <br>Datasheet ↗ | <b>GU1951</b> | "V"         | 4.690m / 15.4ft NAVD88  | Order   | 0.7 mi   |
| <br>Datasheet ↗ | <b>GU1959</b> | "W"         | 21.759m / 71.4ft NAVD88 | Order 1 | 0.8 mi   |
| <br>Datasheet ↗ | <b>GU1949</b> | "RV 1 SPC0" | 3.790m / 12.4ft NAVD88  | Order   | 0.8 mi   |
| <br>Datasheet ↗ | <b>GU1945</b> | "14"        | 5.073m / 16.6ft NAVD88  | Order 1 | 0.8 mi   |

| QR / LINK | PID | DESIGNATION | ELEVATION | ORDER | DISTANCE |
|-----------|-----|-------------|-----------|-------|----------|
| Datasheet |     |             |           |       |          |

FEMA FLOOD ZONE

**X – AREA OF MINIMAL FLOOD HAZARD**

Panel: 06087C0333F

Effective: Sep 29, 2017

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

| PID    | NAME      | LAT       | LON         | ELEV_FT | ORDER |
|--------|-----------|-----------|-------------|---------|-------|
| GU1954 | V         | 36.961142 | -122.041683 | 57.5    | 1     |
| GU1955 | U         | 36.961389 | -122.041389 | 60.9    |       |
| GU4240 | M         | 36.963611 | -122.035556 | 56.7    | 1     |
| GU1950 | U         | 36.963889 | -122.027778 | 11.5    |       |
| GU1951 | V         | 36.963889 | -122.027500 | 15.4    |       |
| GU1959 | W         | 36.959050 | -122.052233 | 71.4    | 1     |
| GU1949 | RV 1 SPC0 | 36.962500 | -122.025000 | 12.4    |       |
| GU1945 | 14        | 36.964722 | -122.026111 | 16.6    | 1     |

Download .txt File

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

| RESERVOIR           | DIST     | ELEVATION           | FULL POOL | STORAGE   | % CAP |
|---------------------|----------|---------------------|-----------|-----------|-------|
| Lake Del Valle      | 47.8 mi  | 696.63 ft (-9 ft)   | 706 ft    | 36K af    | 67%   |
| San Luis Reservoir  | 53.2 mi  | 467 ft (-110 ft)    | 577 ft    | —         | —     |
| Don Pedro Reservoir | 103.3 mi | 811.31 ft (-19 ft)  | 830 ft    | 1,799K af | 89%   |
| Lake McClure        | 106.8 mi | 827.22 ft (-40 ft)  | 867 ft    | 770K af   | 75%   |
| New Melones Lake    | 107.5 mi | 1,019.4 ft (-69 ft) | 1,088 ft  | 1,656K af | 69%   |
| Millerton Lake      | 129.9 mi | 636.59 ft (+59 ft)  | 578 ft    | 53K af    | 10%   |
| Folsom Lake         | 129.9 mi | 446.14 ft (-20 ft)  | 466 ft    | 767K af   | 78%   |

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

| RESERVOIR               | DIST     | ELEVATION | FULL POOL | STORAGE | % CAP |
|-------------------------|----------|-----------|-----------|---------|-------|
| Millerton Lake (Friant) | 128.7 mi | 578 ft*   | 578 ft    | 354K af | 68%   |

ARMY CORPS RESERVOIRS (USACE) – within 150 miles

| RESERVOIR           | DIST   | POOL ELEV | FULL POOL | STORAGE | % CAP |
|---------------------|--------|-----------|-----------|---------|-------|
| Del Valle           | 50 mi  | 696.6 ft  | 695.01 ft | —       | 100%  |
| Farmington Dam-Pool | 101 mi | 124.2 ft  | 117.09 ft | —       | 106%  |
| Camanche            | 113 mi | 230.1 ft  | 221.86 ft | —       | 104%  |
| Burns-Pool          | 125 mi | 265.9 ft  | 267.05 ft | —       | 100%  |
| Bear-Pool           | 127 mi | 344.4 ft  | 344.95 ft | —       | 100%  |
| Eastman Lake-Pool   | 143 mi | 484.8 ft  | 508.76 ft | —       | 95%   |
| Hensley lake-Pool   | 149 mi | 471.6 ft  | 495.29 ft | —       | 95%   |

USGS GROUNDWATER WELLS – within 15 miles

| STATION         | DIST    | WELL DEPTH | WATER LEVEL  | DATE       | AQUIFER    |
|-----------------|---------|------------|--------------|------------|------------|
| 012S001E23R001M | 13.7 mi | 300 ft     | 0.00 ft BGS  | 1975-04-24 | N100CACSTL |
| 012S001E25F001M | 14.2 mi | 604 ft     | 22.40 ft BGS | 1983-09-08 | N100CACSTL |
| 012S001E24Q001M | 14.2 mi | 557 ft     | 24.01 ft BGS | 1983-09-08 | N100CACSTL |
| 012S001E25B003M | 14.3 mi | 580 ft     | 18.72 ft BGS | 1983-08-09 | N100CACSTL |
| 012S001E25G001M | 14.5 mi | 168 ft     | 17.95 ft BGS | 1983-09-08 | N100CACSTL |
| 012S001E25B001M | 14.5 mi | —          | 0.35 ft BGS  | 1983-03-11 | N100CACSTL |
| 012S002E30M002M | 15.1 mi | —          | —            | —          | N100CACSTL |
| 012S002E30N001M | 15.2 mi | —          | —            | —          | N100CACSTL |

○ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The N100CACSTL (Santa Cruz–Santa Lucia Sands and Gravels) aquifer is a Quaternary alluvial and coastal terrace deposit formed by ancient stream erosion and marine reworking of upland sediments during sea-level fluctuations. This permeable sand and gravel unit typically lies within 20 feet of land surface in lowland areas but can extend 600+ feet deep in buried paleochannels. Water availability is moderate to good depending on proximity to active recharge zones and degree of confinement.

FORMATION

|            |                                                                               |
|------------|-------------------------------------------------------------------------------|
| Name       | Santa Cruz–Santa Lucia Sands and Gravels                                      |
| Age / Rock | Quaternary (Holocene–Pleistocene) unconsolidated sand, gravel, silt, and clay |
| Extent     | 800–1200 sq mi (est. USGS/DWR)                                                |

DEPTH & RECHARGE

|               |                         |
|---------------|-------------------------|
| Depth Range   | 0–604 ft (0–60 floors)  |
| Typical Depth | 50–150 ft (5–15 floors) |

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| Recharge Source | Precipitation on coastal plains and foothills + lateral underflow from upland alluvial fans |
| Transit Time    | 5–20 years to 50 ft depth; 20–50 years to 150+ ft depth                                     |

WATER QUALITY

|          |                                                                                        |
|----------|----------------------------------------------------------------------------------------|
| pH       | est. 6.8–7.5 (est. USGS/DWR)                                                           |
| Hardness | est. 150–300 mg/L CaCO3 (est. USGS/DWR)                                                |
| TDS      | est. 250–500 mg/L (est. USGS/DWR)                                                      |
| Notes    | No sample data — typical for formation. Coastal proximity warrants chloride screening. |

TREND

|                |                                                                                                                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trend          | Water levels measured 1975–1983 show fluctuation from near-surface (0.00–0.35 ft BGS in spring/early 1975 and 1983) to 22–24 ft BGS in late summer 1983. Suggests seasonal decline of 18–24 ft over dry season. |
| Source         | USGS NWIS historical well measurements                                                                                                                                                                          |
| Seasonal Swing | 18–24 ft seasonal swing (spring high to late-summer low)                                                                                                                                                        |

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

○ AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The study area near Monterey Bay (36.9588, -122.0384) is situated over the N100CACSTL aquifer system in a region with moderate groundwater resources and no designation as a critically overdrafted basin under SGMA. The somewhat poorly drained soils and coastal hydrogeologic setting have maintained relatively stable aquifer conditions without the intensive agricultural pumping that drives subsidence in California's Central Valley and other over-drafted areas. Surveyors working in this location can proceed with standard elevation surveying practices, as published benchmarks are expected to remain stable and reliable reference points.

SUBSIDENCE ASSESSMENT

|                    |                                    |
|--------------------|------------------------------------|
| Susceptibility     | Low                                |
| Primary Mechanism  | None - consolidated terrain        |
| Annual Rate        | <1 mm/yr (est. USGS)               |
| Historic Loss      | None documented                    |
| Aquifer Compaction | None - aquifer not over-pumped     |
| SGMA Status        | Not applicable - not in SGMA basin |

BENCHMARK IMPLICATIONS

|                     |                                          |
|---------------------|------------------------------------------|
| Monument Risk       | Low - monuments stable                   |
| Elevation Validity  | Good - no active subsidence              |
| GNSS Recommendation | Standard benchmark verification adequate |

SURVEY ACTION

|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| Required Action | No special action required                                               |
| Dataset         | CA DWR SGMA Bulletin 118 2024.1                                          |
| Coverage        | California critically overdrafted basins only - full US coverage pending |

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

|               |             |
|---------------|-------------|
| Site Class    | D (default) |
| Risk Category | II          |

WIND / SNOW / ICE / FROST (ASCE 7-22)

|                                                                |              |
|----------------------------------------------------------------|--------------|
| Basic Wind Speed                                               | 92 mph       |
| Ground Snow Load                                               | 6.4 psf      |
| Winter Wind (W2)                                               | 0.25         |
| Ice Thickness                                                  | None         |
| Frost Depth                                                    | 6 in minimum |
| Brown's Formula · C=1 (Clay, Group D) · AFI=1°F-days · SF=1.25 |              |
| Verify with local building official before construction        |              |

Source: ASCE 7-22 Hazard Tool · gis.asce.org

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Santa Cruz

○ AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate stays mild year-round with minimal temperature swings, so your primary envelope concern is managing moisture and wind rather than extreme heat or cold—prioritize rain-screen details and robust air sealing over heavy insulation. Maximize south and west-facing glazing to capture free solar gain during cooler months, but use deep overhangs or operable shading on those exposures since the marine sun can overheat interior spaces in summer; minimize north-facing glass except where views justify it. The most effective passive strategy here is thermal mass paired with cross-ventilation—use concrete floors or masonry walls to moderate the mild temperature swings, and design operable windows on opposite facades to pull in cool ocean air during afternoons and evenings, reducing mechanical cooling needs significantly.

○ AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal site near Watsonville anticipates moderate wind exposure (92 mph) and light snow loading typical of IECC Zone 3C marine climate. The somewhat poorly drained loam soils and Site Class D classification may warrant a geotechnical investigation to confirm bearing capacity, drainage design, and foundation depth; seismic parameters are currently null and should be verified with local authorities.

FOUNDATION

- Type** Conventional spread footing anticipated, pending geotechnical investigation
- Basis** Seismic design category parameters are null and may warrant confirmation with local building official. Site Class D soils (somewhat poorly drained loam) suggest a geotechnical engineer should assess bearing capacity, settlement potential, and recommended foundation depth.

STRUCTURAL IMPLICATIONS

- Seismic** Seismic design parameters (SDC, Ss, S1, SDS, SD1, PGA) are null for this location. Consult a licensed engineer and local building official to confirm applicable seismic requirements and whether ordinary or special detailing may be warranted.
- Wind Exposure** Exposure C (open terrain with scattered obstructions) anticipated for ASCE 7-22 Risk Category II — typical for coastal development. Site-specific exposure verification recommended during design.
- Wind** Standard wind load design anticipated at 92 mph (3-second gust). Hurricane-force wind design not anticipated for this location.
- Snow** Light snow loading (6.4 psf ground load) anticipated. Standard roof design and drainage likely adequate; consult engineer to confirm roof snow accumulation and wind-driven loading interaction.

INVESTIGATIONS & INSPECTIONS

- Geotech Required** Yes — geotechnical investigation recommended to confirm Site Class D, bearing capacity, drainage characteristics, and settlement potential for the Watsonville loam soils.
- Liquefaction Screen** Liquefaction screening may be warranted once seismic parameters are confirmed. Somewhat poorly drained soils (Hydrologic Group D) and shallow groundwater conditions warrant evaluation by a geotechnical engineer.
- Special Inspections** Special inspections may be required pending confirmation of seismic design category. Consult local building official and structural engineer.
- Code Edition** ASCE 7-22 / IBC 2021 (planning reference only)
- Site Class Note** Site Class D assumed for planning purposes. Geotechnical investigation strongly recommended to confirm soil classification, groundwater depth, and bearing capacity prior to foundation design.

SOILS — USDA SOIL SURVEY

- MAP UNIT:** Watsonville loam, thick surface, 0 to 2 percent slopes      **SERIES:** Watsonville
- DRAINAGE CLASS:** Somewhat poorly drained      **HYDRO GROUP:** D
- SOIL ORDER:** Mollisols      **SUBORDER:** Albolls

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Watsonville loam is a somewhat poorly drained Mollisol (Alboll) that formed in alluvial sediments on nearly level valley floors and low terraces in the Pajaro River basin of central California. The thick surface layer and fine-textured substratum create a seasonally saturated profile with restricted internal drainage; surveyors and developers should expect a high water table in winter and spring, requiring attention to dewatering, fill placement, and foundation design. The nearly flat topography (0–2% slopes) and moderate bearing capacity make this soil suitable for agricultural and light development use, but subsurface moisture conditions and moderate shrink-swell potential demand careful site investigation and drainage planning.

Alluvial parent material on nearly level valley floor and terrace positions; historically grassland and oak savanna vegetation.

DRAINAGE

- Class** Somewhat poorly drained
- Seasonal Saturation** Water table typically 24–36 inches December through April; perched saturation possible in winter
- Flooding Risk** Minimal on 0–2% slopes; site elevation and distance from main channel determine local flood hazard
- Ponding Risk** Low to moderate — gentle slopes minimize ponding, but compacted fill or clay substratum may trap water locally

ENGINEERING

- Bearing Capacity** 2,500–3,500 psf (shallow foundations); reduce by 25–30% if seasonally saturated
- Rating** Fair
- Shrink-Swell** Moderate — fine clay loam substratum expands when wet and shrinks when dry; differential settlement risk
- Frost Action** Low
- Cut Suitability** Fair — fine-grained material with limited gravel; stable cuts to 4–5 feet; monitor for seepage
- Fill Suitability** Fair — clay loam substratum acceptable if properly compacted; moisture control essential to avoid pumping and settlement

SEPTIC

- Suitability** Conditional
- Notes** Perc test required — seasonally high water table and clay loam substratum restrict percolation; design for mounding or raised absorption field; poor long-term performance anticipated

HAZARDS

- Excavation** Low to moderate — fine-grained material excavates easily but may be sticky when wet; seasonal moisture increases excavation difficulty
- Expansive Clay** Moderate — clay-rich substratum exhibits medium expansion potential; foundation design should anticipate differential movement
- Erosion** Low on gentle slopes; gully and surface erosion risk increases if drainage disrupted or subsoil exposed
- Other** Seasonal saturation — dewatering may be required for foundations below 3 feet; verify local groundwater conditions before final design

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision  
Water levels current as of report run: Jul 16, 2026, 2:49 AM UTC  
\* Live pool elevation not reported by this reservoir — full pool design elevation shown

## OIL & GAS WELLS California — within 10mi radius

**13**

total

**0**

active

**13**

plugged

By status:

**13** Plugged

By type: 12 DH 1 CH

| Lease / Well                  | Status  | Type | Operator                      | Depth | Dist    |
|-------------------------------|---------|------|-------------------------------|-------|---------|
| Swanton Improvement Co.       | Plugged | DH   | Swanton Improvement Co.       | —     | 0.9 mi  |
| Danish Oil & Devl Co.         | Plugged | DH   | Danish Oil & Devl Co.         | —     | 3.2 mi  |
| Monterey Bay Oil Co. Ltd      | Plugged | DH   | Monterey Bay Oil Co. Ltd      | —     | 3.3 mi  |
| Greater Stanta Cruz Oil Corp. | Plugged | DH   | Greater Stanta Cruz Oil Corp. | —     | 3.5 mi  |
| Skookum Oil & Invest Corp.    | Plugged | DH   | Skookum Oil & Invest Corp.    | —     | 4.3 mi  |
| Exxon Mobil Corporation       | Plugged | DH   | Exxon Mobil Corporation       | —     | 4.9 mi  |
| Santos, Oneto & Fabry Exp     | Plugged | DH   | Santos, Oneto & Fabry Exp     | —     | 5.6 mi  |
| Santos, Oneto & Fabry Exp     | Plugged | DH   | Santos, Oneto & Fabry Exp     | —     | 5.8 mi  |
| Joseph Faria, Jr.             | Plugged | DH   | Joseph Faria, Jr.             | —     | 6 mi    |
| Donald & Alice Santos         | Plugged | DH   | Donald & Alice Santos         | —     | 6.6 mi  |
| Oneto Bros                    | Plugged | DH   | Oneto Bros                    | —     | 6.9 mi  |
| Shell Western E&P Inc.        | Plugged | CH   | Shell Western E&P Inc.        | —     | 8.2 mi  |
| B. J. Marks Family            | Plugged | DH   | B. J. Marks Family            | —     | 11.5 mi |

### AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is the Swanton Improvement Co. plugged dry hole at 0.9 miles, which is beyond the threshold for actionable review. With no active producers, injection wells, or sealed wells within 500 feet, there are no foundation, environmental, or surface rights issues requiring immediate attention from the well data alone.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200

total in search area

6

within 5 mi

90

within 25 mi

| Site Name                    | Status        | Commodities         | Distance |
|------------------------------|---------------|---------------------|----------|
| Lone Star Industries Inc.    | Past Producer | Limestone, Clay     | 1.7 mi   |
| Unnamed Location             | Occurrence    | LST_D               | 2 mi     |
| Twin Lakes Beach             | Past Producer | Gold                | 2.2 mi   |
| Twin Lakes Beach             | Producer      | Gold                | 2.2 mi   |
| Gibson Sand Fit              | Past Producer | Sand/Gravel         | 3 mi     |
| Unnamed Location             | Occurrence    | Stone/Crushed       | 4.5 mi   |
| Santa Cruz Sand Pit and Mill | Producer      | Sand/Gravel         | 6.2 mi   |
| Decomposed Granite Pit       | Occurrence    | GRT                 | 6.3 mi   |
| Aptos Placer                 | Producer      | Iron, TI_M          | 6.9 mi   |
| Olympia Pit                  | Producer      | Sand/Gravel         | 7.6 mi   |
| Geyer Pit                    | Producer      | Sand/Gravel         | 7.6 mi   |
| Unnamed Location             | Occurrence    | Stone/Crushed       | 7.8 mi   |
| Ixl Quarry                   | Past Producer | LST_D               | 7.9 mi   |
| San Vicente Canyon           | Past Producer | Stone/Crushed, Clay | 9.8 mi   |
| San Vincente Quarry          | Producer      | Stone/Crushed       | 10 mi    |
| Triumph Steel Co             | Occurrence    | Titanium            | 14.5 mi  |
| Alma                         | Prospect      | STN                 | 14.9 mi  |
| Unnamed Location             | Occurrence    | Gold                | 16 mi    |
| Bowie                        | Occurrence    | Mercury             | 16.1 mi  |
| Waddell Creek                | Occurrence    | Gold                | 16.7 mi  |
| Steinheimer Ranch Prospect   | Prospect      | Chromium            | 16.8 mi  |
| Monterey Bay Salt Works      | Producer      | NA_S                | 17.3 mi  |
| Brainard                     | Occurrence    | Mercury             | 17.3 mi  |
| Kilday Ranch Deposits        | Producer      | Chromium            | 17.7 mi  |
| Moss Landing Magnesia Plant  | Plant         | Chromium, MG        | 17.7 mi  |

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

**AI Assessment — Mining Context**

**Active Producers Within 5 Miles:** Twin Lakes Beach (2.2 miles) is an active gold producer. Expect heavy truck traffic on access roads during operating hours and potential noise from mining operations. No other active producers are present within the 5-mile radius.

**Historical Deposits Within 1 Mile:** Lone Star Industries Inc. (1.7 miles) is a past producer of limestone and clay. Proximity warrants consideration of a Phase 1 Environmental Site Assessment given the industrial history.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

**ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi**

6

Endangered

7

Threatened

13

Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

| COMMON NAME                     | SCIENTIFIC NAME                                  | STATUS     |
|---------------------------------|--------------------------------------------------|------------|
| Robust spineflower              | <i>Chorizanthe robusta</i> var. <i>robusta</i>   | Endangered |
| Scotts Valley polygonum         | <i>Polygonum hickmanii</i>                       | Endangered |
| Scotts Valley spineflower       | <i>Chorizanthe robusta</i> var. <i>hartwegii</i> | Endangered |
| tidewater goby                  | <i>Eucyclogobius newberryi</i>                   | Endangered |
| Yadon's piperia                 | <i>Piperia yadonii</i>                           | Endangered |
| Zayante band-winged grasshopper | <i>Trimerotropis infantilis</i>                  | Endangered |
| Bay checkerspot butterfly       | <i>Euphydryas editha bayensis</i>                | Threatened |
| California red-legged frog      | <i>Rana draytonii</i>                            | Threatened |
| California tiger Salamander     | <i>Ambystoma californiense</i>                   | Threatened |
| Marbled Murrelet                | <i>Brachyramphus marmoratus</i>                  | Threatened |
| Monterey spineflower            | <i>Chorizanthe pungens</i> var. <i>pungens</i>   | Threatened |
| Santa Cruz tarplant             | <i>Holocarpha macradenia</i>                     | Threatened |
| western snowy plover            | <i>Charadrius nivosus nivosus</i>                | Threatened |

**AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU**

ESA Implications Review – 36.9588, -122.0384 **Section 7 Trigger & Key Species:** If the proposed project involves federal permitting, funding, or licensing, ESA Section 7 consultation with USFWS/NOAA is mandatory given the 13 species with critical habitat within 25 miles. For planning purposes, the most project-relevant species likely include California red-legged frog and California tiger salamander (if wetlands/seasonal pools are present), tidewater goby (if riparian/coastal access exists), and Marbled Murrelet (if coast redwood habitat is on-site). The terrestrial spineflower and grasshopper species warrant attention if grassland or coastal sage scrub habitats occupy the project footprint. **Recommended Next Steps:** Conduct a species-specific habitat assessment to determine actual presence/absence and critical habitat overlap on the property, then initiate early coordination with USFWS Sacramento Field Office to clarify Section 7 requirements and identify avoidance/minimization measures before final project design. This proactive approach can substantially reduce permitting delays and costs. --- \*For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.\*

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

**NATIVE PLANTS - iNaturalist Research-Grade within 10mi**

| 52                            | 51800                                 | 2026-07-15  |
|-------------------------------|---------------------------------------|-------------|
| Unique Species                | Total Observations                    | Most Recent |
| COMMON NAME                   | SCIENTIFIC NAME                       | LAST OBS    |
| American lyme grass           | <i>Leymus mollis</i>                  | 2026-07-13  |
| American trailplant           | <i>Adenocaulon bicolor</i>            | 2026-07-09  |
| Arroyo Willow                 | <i>Salix lasiolepis</i>               | 2026-07-13  |
| beach suncup                  | <i>Camissoniopsis cheiranthifolia</i> | 2026-07-13  |
| beaked hazelnut               | <i>Corylus cornuta</i>                | 2026-07-09  |
| bigleaf maple                 | <i>Acer macrophyllum</i>              | 2026-07-09  |
| bishop pine                   | <i>Pinus muricata</i>                 | 2026-07-14  |
| California bay                | <i>Umbellularia californica</i>       | 2026-07-12  |
| California fetid adderstongue | <i>Scoliopus bigelovii</i>            | 2026-07-09  |
| California Harebell           | <i>Smithiastrum prenanthoides</i>     | 2026-07-08  |
| California Hedge Nettle       | <i>Stachys bullata</i>                | 2026-07-10  |
| California incense-cedar      | <i>Calocedrus decurrens</i>           | 2026-07-15  |
| California mugwort            | <i>Artemisia douglasiana</i>          | 2026-07-08  |
| California poppy              | <i>Eschscholzia californica</i>       | 2026-07-13  |
| California Spikenard          | <i>Aralia californica</i>             | 2026-07-08  |
| California torreyia           | <i>Torreya californica</i>            | 2026-07-08  |
| coast live oak                | <i>Quercus agrifolia</i>              | 2026-07-13  |
| coast redwood                 | <i>Sequoia sempervirens</i>           | 2026-07-13  |
| coastal bush lupine           | <i>Lupinus arboreus</i>               | 2026-07-13  |
| coffeeberry                   | <i>Frangula californica</i>           | 2026-07-11  |
| common selfheal               | <i>Prunella vulgaris</i>              | 2026-07-15  |
| creeping snowberry            | <i>Symphoricarpos mollis</i>          | 2026-07-13  |
| Evergreen Huckleberry         | <i>Vaccinium ovatum</i>               | 2026-07-09  |
| fragrant bedstraw             | <i>Galium triflorum</i>               | 2026-07-13  |
| Green pin-cushion alga        | <i>Cladophora columbiana</i>          | 2026-07-12  |
| lady fern                     | <i>Athyrium filix-femina</i>          | 2026-07-09  |
| Monterey Monkeyflower         | <i>Diplacis linearis</i>              | 2026-07-09  |
| Narrowleaf Willow             | <i>Salix exigua</i>                   | 2026-07-13  |

| COMMON NAME              | SCIENTIFIC NAME                     | LAST OBS   |
|--------------------------|-------------------------------------|------------|
| orange bush monkeyflower | <i>Diplacus aurantiacus</i>         | 2026-07-12 |
| Pacific False Bindweed   | <i>Calystegia purpurata</i>         | 2026-07-10 |
| Pacific hound's tongue   | <i>Adelinia grandis</i>             | 2026-07-13 |
| Pacific madrone          | <i>Arbutus menziesii</i>            | 2026-07-15 |
| Pacific poison oak       | <i>Toxicodendron diversilobum</i>   | 2026-07-13 |
| Pacific trillium         | <i>Trillium ovatum</i>              | 2026-07-09 |
| Pink Honeysuckle         | <i>Lonicera hispidula</i>           | 2026-07-08 |
| Pink Sand Verbena        | <i>Abronia umbellata</i>            | 2026-07-12 |
| Redwood Sorrel           | <i>Oxalis smalliana</i>             | 2026-07-12 |
| Redwood Violet           | <i>Viola sempervirens</i>           | 2026-07-09 |
| Scouring-pad alga        | <i>Endocladia muricata</i>          | 2026-07-12 |
| Seaside Buckwheat        | <i>Eriogonum latifolium</i>         | 2026-07-14 |
| seep monkeyflower        | <i>Erythranthe guttata</i>          | 2026-07-09 |
| Stream Orchid            | <i>Epipactis gigantea</i>           | 2026-07-08 |
| tall flatsedge           | <i>Cyperus eragrostis</i>           | 2026-07-09 |
| Tanoak                   | <i>Notholithocarpus densiflorus</i> | 2026-07-09 |
| thimbleberry             | <i>Rubus parviflorus</i>            | 2026-07-09 |
| tomcat clover            | <i>Trifolium willdenovii</i>        | 2026-07-13 |
| western blue-eyed grass  | <i>Sisyrinchium bellum</i>          | 2026-07-09 |
| western maidenhair fern  | <i>Adiantum aleuticum</i>           | 2026-07-09 |
| western sword fern       | <i>Polystichum munitum</i>          | 2026-07-12 |
| Western Wild Ginger      | <i>Asarum caudatum</i>              | 2026-07-09 |
| wing-leaf monkeyflower   | <i>Erythranthe pilota</i>           | 2026-07-09 |
| yerba buena              | <i>Clinopodium douglasii</i>        | 2026-07-15 |

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 36.9588, -122.0384 This site supports a **coastal mixed hardwood and conifer forest community**, characteristic of the California Central Coast transitional zone between redwood and oak woodlands. The species composition—dominated by coast redwoods, bishop pines, coast live oaks, and coastal understory species like California bay and beaked hazelnut—indicates a cool, moisture-rich environment with moderate to high rainfall and marine influence. Soil conditions are likely well-drained but consistently moist due to fog drip and seasonal precipitation, with significant organic matter accumulation in the understory; grading activities should account for shallow root systems in redwoods and poor drainage in compacted soils, and drainage design should preserve natural water infiltration patterns rather than accelerate runoff that could destabilize slopes. The presence of moisture-dependent species (California spikenard, arroyo willow, redwood sorrel associates) indicates seasonal water retention, requiring careful assessment of cut/fill impacts and potential for erosion or nutrient leaching post-development. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

|                            |                  |                  |              |
|----------------------------|------------------|------------------|--------------|
| 242                        | 199              | 150              | 3092         |
| Venomous Snakes            | Venomous Spiders | Stinging Insects | Toxic Plants |
| VENOMOUS SNAKES            |                  |                  |              |
| SPECIES                    |                  | OBSERVATIONS     |              |
| Crotalus oreganus oreganus |                  | 242 obs          |              |
| Crotalus oreganus          |                  | 242 obs          |              |
| VENOMOUS SPIDERS           |                  |                  |              |
| SPECIES                    |                  | OBSERVATIONS     |              |
| Latrodectus hesperus       |                  | 199 obs          |              |
| Latrodectus geometricus    |                  | 199 obs          |              |
| STINGING INSECTS           |                  |                  |              |
| SPECIES                    |                  | OBSERVATIONS     |              |
| Dasymutilla sackenii       |                  | 150 obs          |              |
| Dasymutilla aureola        |                  | 150 obs          |              |
| Dasymutilla californica    |                  | 150 obs          |              |
| Dasymutilla flammifera     |                  | 150 obs          |              |
| TOXIC PLANTS               |                  |                  |              |
| SPECIES                    |                  | OBSERVATIONS     |              |
| Toxicodendron diversilobum |                  | 1967 obs         |              |

| SPECIES                            | OBSERVATIONS |
|------------------------------------|--------------|
| <i>Datura stramonium</i>           | 86 obs       |
| <i>Brugmansia aurea</i>            | 86 obs       |
| <i>Datura wrightii</i>             | 86 obs       |
| <i>Datura discolor</i>             | 86 obs       |
| <i>Conium maculatum</i>            | 889 obs      |
| <i>Urtica gracilis</i>             | 150 obs      |
| <i>Urtica dioica</i>               | 150 obs      |
| <i>Urtica gracilis holosericea</i> | 150 obs      |

**AI FIELD SAFETY BRIEFING - CLAUDE HAIKU**

Field Safety Brief – Santa Cruz County Area **Highest-Priority Hazards:** Toxic plants dominate this region with nearly 3,100 observations—particularly poison oak (\*Toxicodendron diversilobum\*), which causes severe skin reactions on contact and is present year-round. Northern Pacific rattlesnakes (\*Crotalus oreganus oreganus\*) are common and pose a serious bite risk; black widow spiders are also present. Stinging insects, including velvet ants (wingless wasps), can deliver painful stings. **Practical Avoidance:** Wear long pants, long sleeves, and closed-toe boots; wash clothing immediately after fieldwork to remove plant oils. Watch for coiled snakes near brush and rock; use a walking stick to create vibrations before stepping into tall vegetation. Learn to identify poison oak by its three-leaflet pattern and avoid touching any unfamiliar plants. Stay alert during brush work and never place hands or feet where you cannot see clearly. **Seasonal Note:** Snakes are most active April–October; poison oak remains hazardous year-round but causes more acute reactions during active growth (spring through fall). Stinging insects peak in warmer months. **Always verify current conditions locally before fieldwork.**

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FAA AIRWAYS — ATS Routes crossing site area

| 19            | 12            | 3              | 4            |          |        |           |
|---------------|---------------|----------------|--------------|----------|--------|-----------|
| Total Airways | Jet Routes    | Victor Airways | Other Routes |          |        |           |
| AIRWAY        | TYPE          | LEVEL          | MEA EAST     | MEA WEST | TRACK  | REV TRACK |
| J1            | Jet Route     | High           | 18,000 ft    | —        | 303.8° | 121.4°    |
| J110          | Jet Route     | High           | 18,000 ft    | —        | 137.9° | 318.2°    |
| J126          | Jet Route     | High           | 18,000 ft    | —        | 308.1° | 127.9°    |
| J3            | Jet Route     | High           | 18,000 ft    | —        | 342.8° | 161.8°    |
| J32           | Jet Route     | High           | 18,000 ft    | —        | 19.2°  | 199.6°    |
| J501          | Jet Route     | High           | 18,000 ft    | —        | 301.5° | 121.1°    |
| J58           | Jet Route     | High           | 18,000 ft    | —        | 65.3°  | 245.9°    |
| J6            | Jet Route     | High           | 18,000 ft    | —        | 127.8° | 307.9°    |
| J80           | Jet Route     | High           | 18,000 ft    | —        | 65.3°  | 245.9°    |
| J84           | Jet Route     | High           | 18,000 ft    | —        | 50.8°  | 231.0°    |
| J88           | Jet Route     | High           | 18,000 ft    | —        | 308.1° | 127.9°    |
| J94           | Jet Route     | High           | 18,000 ft    | —        | 65.3°  | 245.9°    |
| T257          | T-Route       | Low            | —            | —        | 322.0° | 141.7°    |
| T259          | T-Route       | Low            | —            | —        | 41.6°  | 221.8°    |
| T261          | T-Route       | Low            | —            | —        | 31.4°  | 211.7°    |
| T263          | T-Route       | Low            | —            | —        | 308.8° | 128.2°    |
| V107          | Victor Airway | Low            | 7,000 ft     | —        | 295.4° | 114.2°    |
| V109          | Victor Airway | Low            | 4,000 ft     | —        | 240.1° | 60.0°     |
| V111          | Victor Airway | Low            | 5,500 ft     | —        | 11.0°  | 191.0°    |

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

| 7                |          |        |           |          |                     |  |
|------------------|----------|--------|-----------|----------|---------------------|--|
| Rail Lines       |          |        |           |          |                     |  |
| OWNER / OPERATOR | TYPE     | TRACKS | PASSENGER | STRACNET | SUBDIVISION         |  |
| SCBG             | X        | 1      | No        | No       | SAN FRANCISCO       |  |
| SPP              | X        | 1      | No        | No       | COASTAL SUBDIVISION |  |
| SCBG             | Mainline | 1      | No        | No       | COASTAL SUBDIVISION |  |
| SCBG             | Other    | 1      | No        | No       | —                   |  |
| SCBG             | S        | 1      | No        | No       | —                   |  |
| SPP              | Mainline | 1      | No        | No       | COASTAL SUBDIVISION |  |
| Unknown          | A        | 1      | No        | No       | —                   |  |

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

|                    |               |       |         |          |            |
|--------------------|---------------|-------|---------|----------|------------|
| 200                | 88            | 64    | 13      | 10       | 1          |
| Total Observations | Total Species | Birds | Mammals | Reptiles | Amphibians |

BIRDS (64 species)

| COMMON NAME          | Scientific Name         | OBS |
|----------------------|-------------------------|-----|
| Rock Pigeon          | Columba livia           | 12  |
| Feral Pigeon         | Columba livia domestica | 12  |
| Western Gull         | Larus occidentalis      | 8   |
| Wild Turkey          | Meleagris gallopavo     | 7   |
| California Towhee    | Melospiza crissalis     | 4   |
| Wood Duck            | Aix sponsa              | 4   |
| Great Horned Owl     | Bubo virginianus        | 3   |
| Turkey Vulture       | Cathartes aura          | 3   |
| Heermann's Gull      | Larus heermanni         | 3   |
| Cooper's Hawk        | Astur cooperii          | 3   |
| California Gull      | Larus californicus      | 3   |
| Red-necked Phalarope | Phalaropus lobatus      | 3   |
| Common Murre         | Uria aalge              | 3   |
| Pigeon Guillemot     | Cephus columba          | 3   |
| California Quail     | Callipepla californica  | 2   |

+ 49 additional species

MAMMALS (13 species)

| COMMON NAME                                | Scientific Name                 | OBS |
|--------------------------------------------|---------------------------------|-----|
| Eastern Pacific Long-beaked Common Dolphin | Delphinus delphis bairdii       | 14  |
| Humpback Whale                             | Megaptera novaeangliae          | 8   |
| California Ground Squirrel                 | Otospermophilus beecheyi        | 6   |
| Common Bottlenose Dolphin                  | Tursiops truncatus              | 3   |
| California Sea Lion                        | Zalophus californianus          | 2   |
| Mule Deer                                  | Odocoileus hemionus             | 2   |
| Columbian Black-tailed Deer                | Odocoileus hemionus columbianus | 2   |
| Coyote                                     | Canis latrans                   | 2   |
| Harbor Seal                                | Phoca vitulina                  | 2   |
| Southern Sea Otter                         | Enhydra lutris nereis           | 1   |
| Eastern Gray Squirrel                      | Sciurus carolinensis            | 1   |
| Pacific Harbor Seal                        | Phoca vitulina richardii        | 1   |
| Long-tailed Weasel                         | Neogale frenata                 | 1   |

REPTILES (10 species)

| COMMON NAME                     | Scientific Name                   | OBS |
|---------------------------------|-----------------------------------|-----|
| Western Fence Lizard            | Sceloporus occidentalis           | 6   |
| Coast Garter Snake              | Thamnophis elegans terrestris     | 4   |
| Pacific Gopher Snake            | Pituophis catenifer catenifer     | 2   |
| Santa Cruz Aquatic Garter Snake | Thamnophis atratus atratus        | 1   |
| California King Snake           | Lampropeltis californiae          | 1   |
| Western Yellow-bellied Racer    | Coluber constrictor mormon        | 1   |
| Northern Alligator Lizard       | Elgaria coerulea                  | 1   |
| Forest Sharp-tailed Snake       | Contia longicauda                 | 1   |
| Southern Alligator Lizard       | Elgaria multicarinata             | 1   |
| Coast Range Fence Lizard        | Sceloporus occidentalis bocourtii | 1   |

AMPHIBIANS (1 species)

| COMMON NAME         | Scientific Name  | OBS |
|---------------------|------------------|-----|
| Arboreal Salamander | Aneides lugubris | 1   |

Ecological Assessment: 36.9588, -122.0384 This site, located in the central California coast region, supports a diverse wildlife community indicative of mixed coastal scrub, grassland, and riparian habitats with significant marine influence. The presence of marine mammals (dolphins, whales, sea otters, seals) and marine/coastal birds (gulls, cormorants) confirms proximity to productive ocean waters, while terrestrial species (squirrels, deer, coyotes, raptors) reflect upland and transitional habitats. Multiple species of regulatory concern are present, including the Southern Sea Otter (federally listed threatened), coastal raptors (Great Horned Owl, Cooper's Hawk), and specialized aquatic species (Santa Cruz Aquatic Garter Snake, Heermann's Gull), suggesting sensitive riparian corridors, coastal bluffs, and nearshore marine zones requiring resource protection. The wildlife assemblage—characterized by both generalist species (pigeons, turkeys, coyotes) and sensitive natives—indicates a mosaic of developed/disturbed areas adjacent to relatively intact natural communities, with moderate to high habitat connectivity to marine and terrestrial reserves. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 36.96°

1 • AVERAGE SUN HOURS

| Period            | Peak Hrs/Day | Notes                        |
|-------------------|--------------|------------------------------|
| Annual average    | 6.7 hrs      | Overall solar resource       |
| Summer average    | 9.1 hrs      | June–August                  |
| Winter average    | 4 hrs        | December–February            |
| Worst month (Dec) | 3.3 hrs      | Size battery storage to this |

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 37° (latitude) → 9/12 pitch

| Pitch | Angle | % of Optimal | Notes            |
|-------|-------|--------------|------------------|
| 4/12  | 18.4° | 97.2%        | good             |
| 5/12  | 22.6° | 97.8%        | good             |
| 6/12  | 26.6° | 98.4%        | ↔ excellent      |
| 7/12  | 30.3° | 99%          | ↔ excellent      |
| 8/12  | 33.7° | 99.5%        | ↔ excellent      |
| 9/12  | 36.9° | 100%         | ↔ annual optimal |
| 10/12 | 39.8° | 99.6%        | ↔ excellent      |
| 12/12 | 45°   | 98.8%        | ↔ excellent      |

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +734 kWh/yr (~\$110 value/yr). Over 25yr panel life: ~\$2751 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

| Surface                    | Winter | Summer | Design Note                          |
|----------------------------|--------|--------|--------------------------------------|
| South glass (SHGC 0.4)     | 54     | 31     | Minimize south glass                 |
| South glass (SHGC 0.6)     | 81     | 46     | Clear/uncoated glass                 |
| North glass (any)          | 10     | 13     | Negligible — diffuse only            |
| East / West glass          | 29     | 105    | ⚠ Minimize — afternoon load          |
| Roof — dark (absorb 0.95)  | 241    | 301    | Highest load surface                 |
| Roof — light (absorb 0.35) | 89     | 111    | Cool roof — significant reduction    |
| Roof — metal (absorb 0.25) | 63     | 79     | Best roof option for heat            |
| South wall — dark          | 100    | 69     |                                      |
| South wall — light         | 33     | 23     | Light color = major savings          |
| Hardscape — asphalt        | 181    | 301    | ⚠ Heat island — avoid near structure |
| Hardscape — concrete       | 124    | 206    | Also reflects onto adjacent walls    |
| Hardscape — light pave     | 67     | 111    | Recommended within 50ft              |
| Pool surface               | 23     | 38     | Low absorb + evaporative cooling     |

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.

Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

| Pitch | Dark Roof | Light Roof | Reduction vs Flat      |
|-------|-----------|------------|------------------------|
| 0/12  | 301 BTU   | 111 BTU    | baseline (worst)       |
| 4/12  | 241 BTU   | 89 BTU     | -20% dark / -20% light |
| 6/12  | 196 BTU   | 72 BTU     | -35% dark / -35% light |
| 8/12  | 166 BTU   | 61 BTU     | -45% dark / -45% light |
| 12/12 | 135 BTU   | 50 BTU     | -55% dark / -55% light |

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

| Assembly          | Lag     | Peak Interior Load | Note                                       |
|-------------------|---------|--------------------|--------------------------------------------|
| 2x4 frame (R-15)  | 0.6 hrs | ~15:36 PM          | R-value only — no mass benefit             |
| 2x6 frame (R-23)  | 0.9 hrs | ~15:54 PM          | R-value only — no mass benefit             |
| 2x8 frame (R-30)  | 1.2 hrs | ~16:12 PM          | R-value only — no mass benefit             |
| 2x10 frame (R-37) | 1.5 hrs | ~16:30 PM          | good                                       |
| 2x12 frame (R-44) | 1.8 hrs | ~16:48 PM          | good                                       |
| ICF 6" core       | 7 hrs   | ~22:00 PM          | ✓ shifts past sunset — natural vent window |
| 8" CMU block      | 6 hrs   | ~21:00 PM          | ✓ shifts past sunset — natural vent window |
| 12" concrete      | 8.5 hrs | ~23:30 PM          | ✓ shifts past sunset — natural vent window |

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

| Tree Ht | Shadow Reaches | Summer hrs/day | Equinox hrs/day | Winter hrs/day | WUI note |
|---------|----------------|----------------|-----------------|----------------|----------|
| 10ft    | All year       | 1.1            | 1               | 1.1            | OK       |
| 20ft    | All year       | 2.1            | 1.9             | 2.3            | OK       |

|      |          |     |     |     |              |
|------|----------|-----|-----|-----|--------------|
| 30ft | All year | 3   | 2.8 | 3.6 | Zone 2 check |
| 40ft | All year | 3.9 | 3.7 | 5   | △ verify     |
| 50ft | All year | 4.7 | 4.5 | 6.7 | △ verify     |
| 60ft | All year | 5.5 | 5.3 | 9.5 | ✗ Zone 1     |

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.  
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.  
△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

| Standard        | Unbalanced (1/150) | Balanced† (1/300) | Notes                                 |
|-----------------|--------------------|-------------------|---------------------------------------|
| Code minimum    | 8 sq in            | 4 sq in           | Moisture protection only              |
| Thermal comfort | 24 sq in           | 12 sq in          | Keeps attic within 20°F of outdoor    |
| WUI mesh 1/16"  | 60 sq in           | 30 sq in          | 2.5× compensation — mesh clogs        |
| WUI intumescent | 24 sq in           | 12 sq in          | Self-closing — no compensation needed |

**Pitch factor:** 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45  
**Color factor:** Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45  
**Scale:** multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor  
†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)  
**WUI conflict:** Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.  
△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🕒 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Solar & Energy Planning for Your Home Your location gets about 6.7 hours of useful sunlight per day on average, which is solid for solar panels—think of it like having a reliable 4 PM to 10 PM work shift, seven days a week. A south-facing roof tilted at roughly a 37-degree angle (almost a 9-on-12 pitch) will catch that sun most efficiently, but here's where it gets interesting: if your roof is dark, it'll absorb summer heat like a black car in the sun and dump 301 BTU of unwanted warmth into your home every hour per square foot—but switching to a light color cuts that nearly in half. Your west-facing windows are actually a bigger problem in summer than most people expect, pulling in 105 BTU per hour as that afternoon sun blasts through, so you'll want to think seriously about shading or reflective glass on that side. This is exactly why planting a large deciduous tree on the south side is so smart: it blocks harsh summer sun for about 4 hours but drops its leaves in winter and lets warming sun through when you need it. One more useful concept—thermal mass just means how well your walls slow down temperature swings, so thicker, denser walls hold heat longer and keep your home steadier. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING 36.95882°N, 122.03836°W | Santa Cruz County, CA

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### Structural Design Drivers

This marine-influenced IECC Zone 3C site exhibits neither documented snow nor wind hazard data in the record, eliminating those parameters from the primary structural envelope. The absence of seismic data precludes quantification of ground motion demand, but Santa Cruz County's documented seismic activity and proximity to the San Andreas system warrant application of CBC seismic provisions as baseline; absence of data does not indicate absence of hazard. Design team must source site-specific seismic evaluation independently. The structural system will be governed by local building code defaults for Zone 3C marine climate rather than by extreme environmental loading.

### Geotechnical & Foundation Considerations

Watsonville loam (Hydro Group D, somewhat poorly drained) presents the controlling geotechnical constraint. Group D classification indicates high runoff potential and poor infiltration; combined with stated "somewhat poorly drained" surface conditions, this soil requires careful moisture management and likely precludes shallow spread footings without subsurface investigation and possible mitigation. The 6-inch frost depth (derived from Brown's Formula with AFI of 1°F-days) is relatively shallow and non-controlling for standard footing placement; however, the low-permeability, moisture-retentive character of loam in Group D means frost heave risk is present despite nominal depth and must be evaluated through site-specific soils boring. Recommend SPT or CPT to establish bearing capacity, seasonal water table elevation, and consolidation characteristics before foundation type selection.

### Hazard Assessment

No Quaternary faults are recorded within the search radius, and no subsidence basin data flags the site. Wind and snow hazards carry no recorded values. Wildfire data are not provided in this dataset. The seismic hazard remains unquantified by the data supplied; independent USGS or site-specific seismic hazard evaluation is essential before final structural design.

### Site Context & Mining Proximity

Twin Lakes Beach gold producer operates 2.2 miles distant; 200 additional MRDS deposits lie within ~35 miles. No historical or active subsidence or ground instability from mining is documented at this location. Thirteen oil and gas wells exist within ~10 miles, but all are either plugged or inactive (nearest: Swanton Improvement Co., 0.9 miles, plugged). No active hydrocarbon extraction presents a constraint. The site's marine climate envelope (Zone 3C, AFI 1°F-days) aligns with coastal Santa Cruz County conditions and does not impose exceptional building envelope or durability demands beyond standard marine exposure protocols.

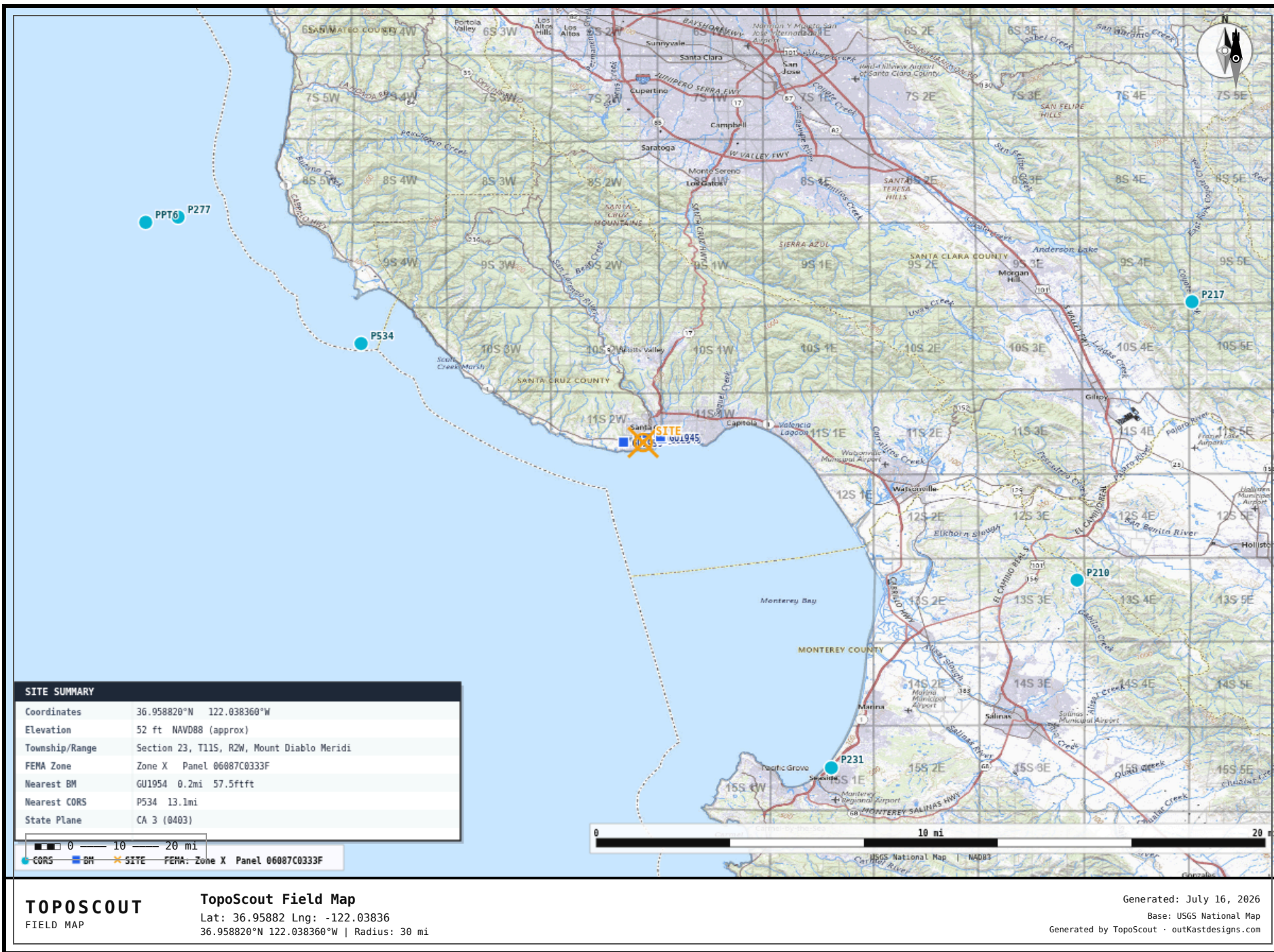
### Bottom Line

Initiate subsurface soils investigation (boring/SPT) to establish bearing capacity, water table, and frost/heave risk given Watsonville loam's Group D drainage class, and obtain site-specific seismic hazard assessment before finalizing structural system and foundation design.

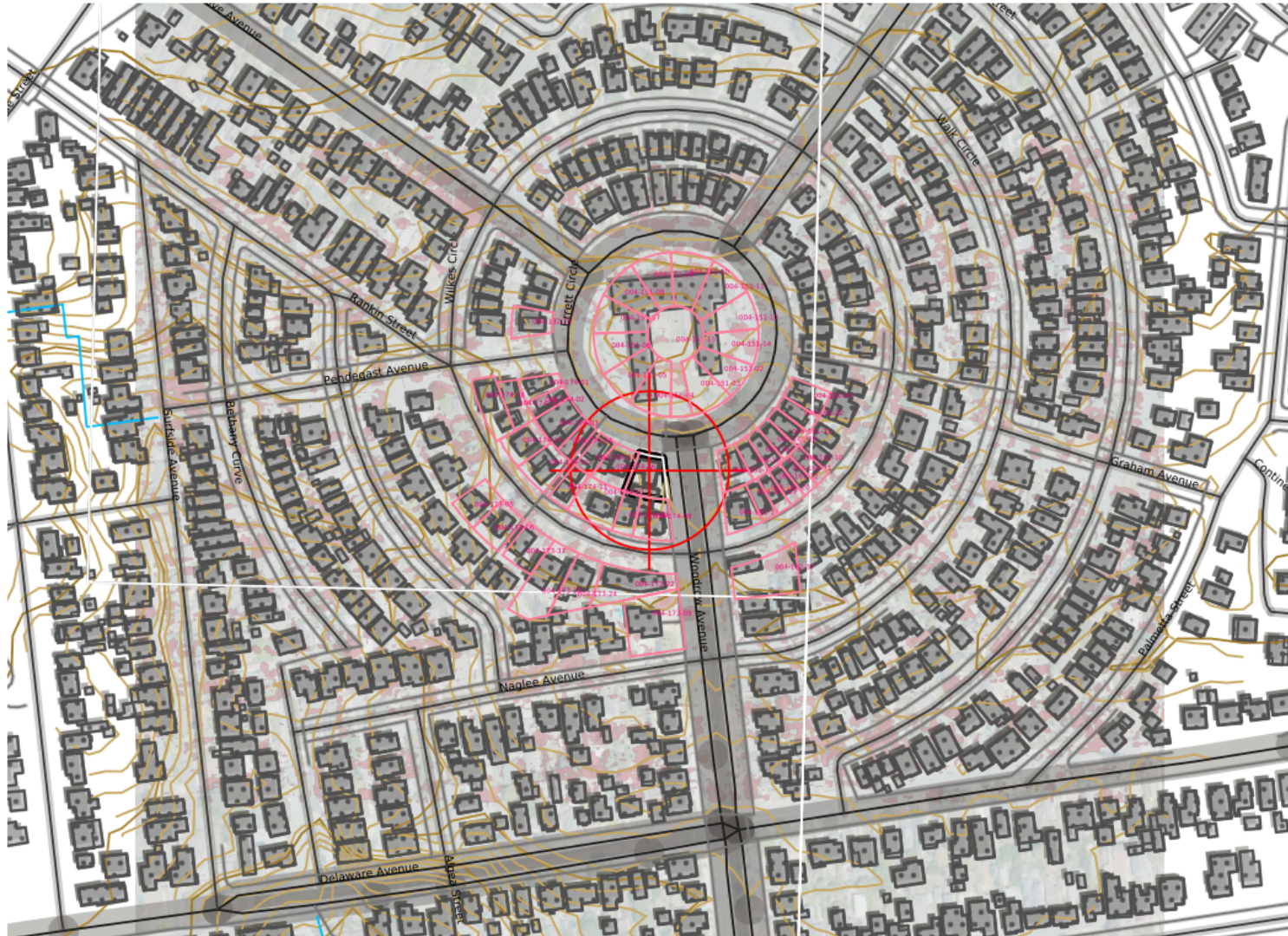


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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

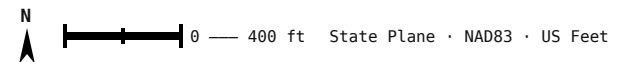


## TOPOSCOUT SITE PLAN PREVIEW



### INCLUDED LAYERS

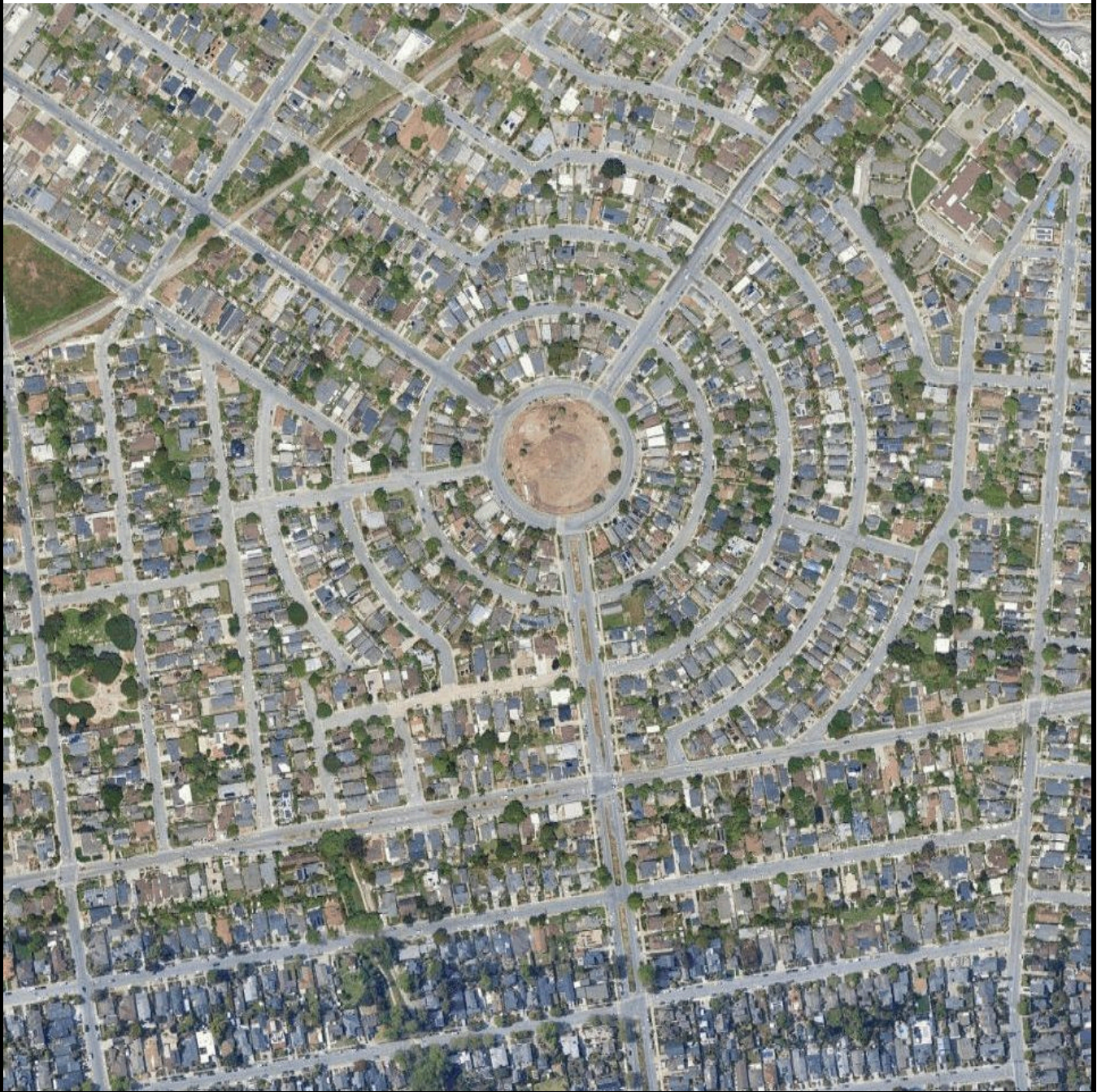
- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polyline (all geometry)





**SITE AERIAL – 600 ft × 600 ft**

36.95882°N, 122.03836°W · Google Aerial



**AREA CONTEXT — 3000 ft × 3000 ft**

36.95882°N, 122.03836°W · Google Aerial

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## ✂ FIELD RESOURCE LINKS

### PRE-MISSION PLANNING

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#### Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

#### FAA B4UFLY –

[https://www.faa.gov/uas/recreational\\_flyers/where\\_can\\_i\\_fly/b4ufly](https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly)

Drone airspace authorization, TFRs, restricted areas

#### Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

#### NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

### POST-PROCESSING

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#### NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

#### NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

#### CORS Data Download – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Raw RINEX observation files from CORS stations

### FEMA & FLOOD

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#### FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

#### FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

### BASE STATION & NETWORK

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#### NOAA CORS Network – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Continuously operating reference stations, data download

#### NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

#### IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

### REFERENCE & CALCULATIONS

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#### NGS Datasheets – <https://www.ngs.noaa.gov/cgi-bin/datasheet.prl>

Benchmarks, control points, passive monuments

#### NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

#### NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

#### NGS Geoid Models – [https://geodesy.noaa.gov/GEOID/](https://geodesy.noaa.gov/GEOID/GEOID18)

GEOID18 and current model downloads, GEOID calculator

### DRONE OPERATIONS

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#### FAA Part 107 — Commercial Rules –

[https://www.faa.gov/uas/commercial\\_operators](https://www.faa.gov/uas/commercial_operators)

Full Part 107 regulatory summary for commercial UAS operations

#### FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

#### FAA B4UFLY –

[https://www.faa.gov/uas/recreational\\_flyers/where\\_can\\_i\\_fly/b4ufly](https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly)

Pre-flight airspace check, TFRs, LAANC authorization

#### FAA Part 107 Waivers –

[https://www.faa.gov/uas/commercial\\_operators/part\\_107\\_waivers](https://www.faa.gov/uas/commercial_operators/part_107_waivers)

Waiver applications for operations beyond standard 107 rules

#### Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

### AIR QUALITY & WILDFIRE SMOKE

#### [AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

#### [AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

#### [NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

